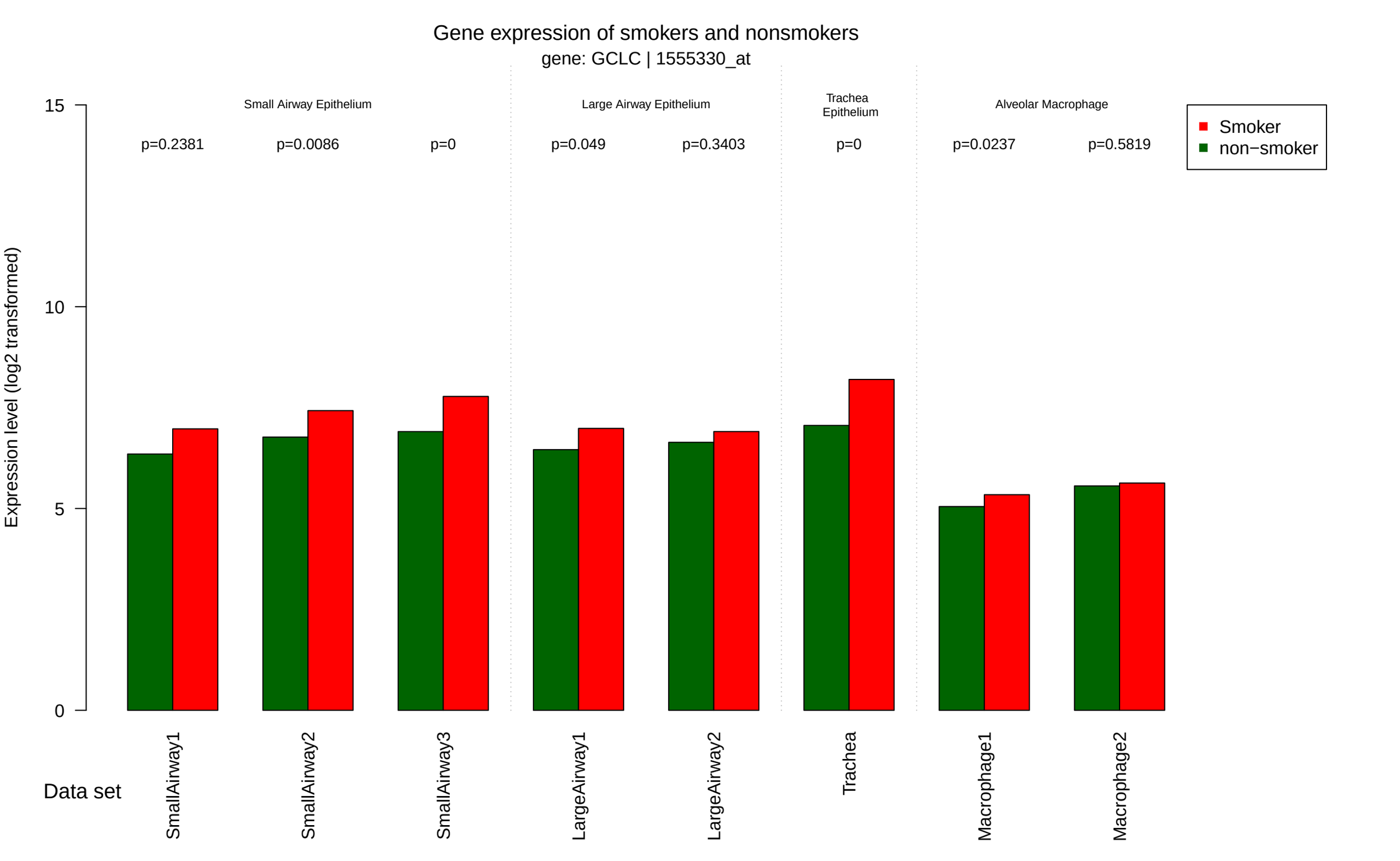
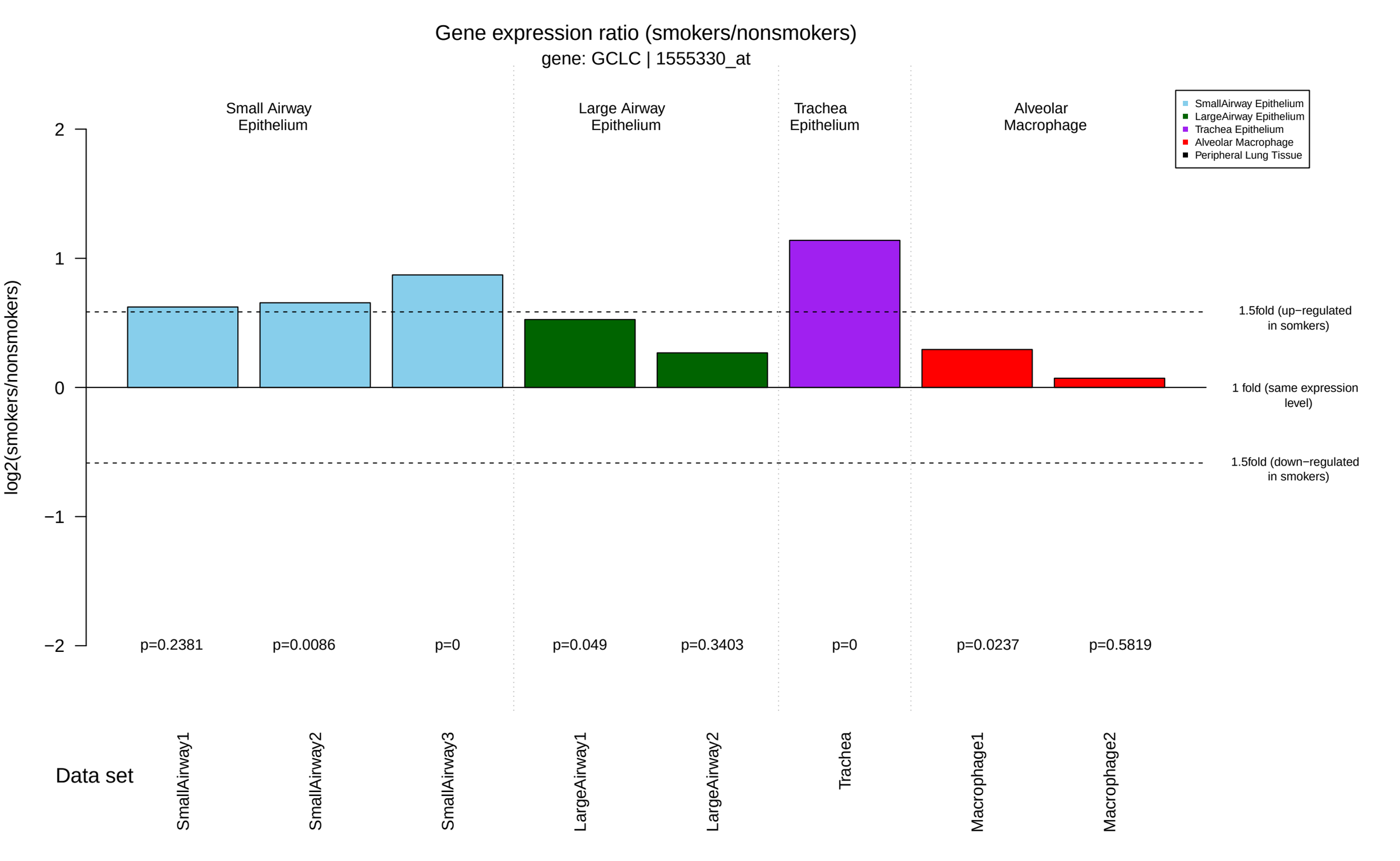
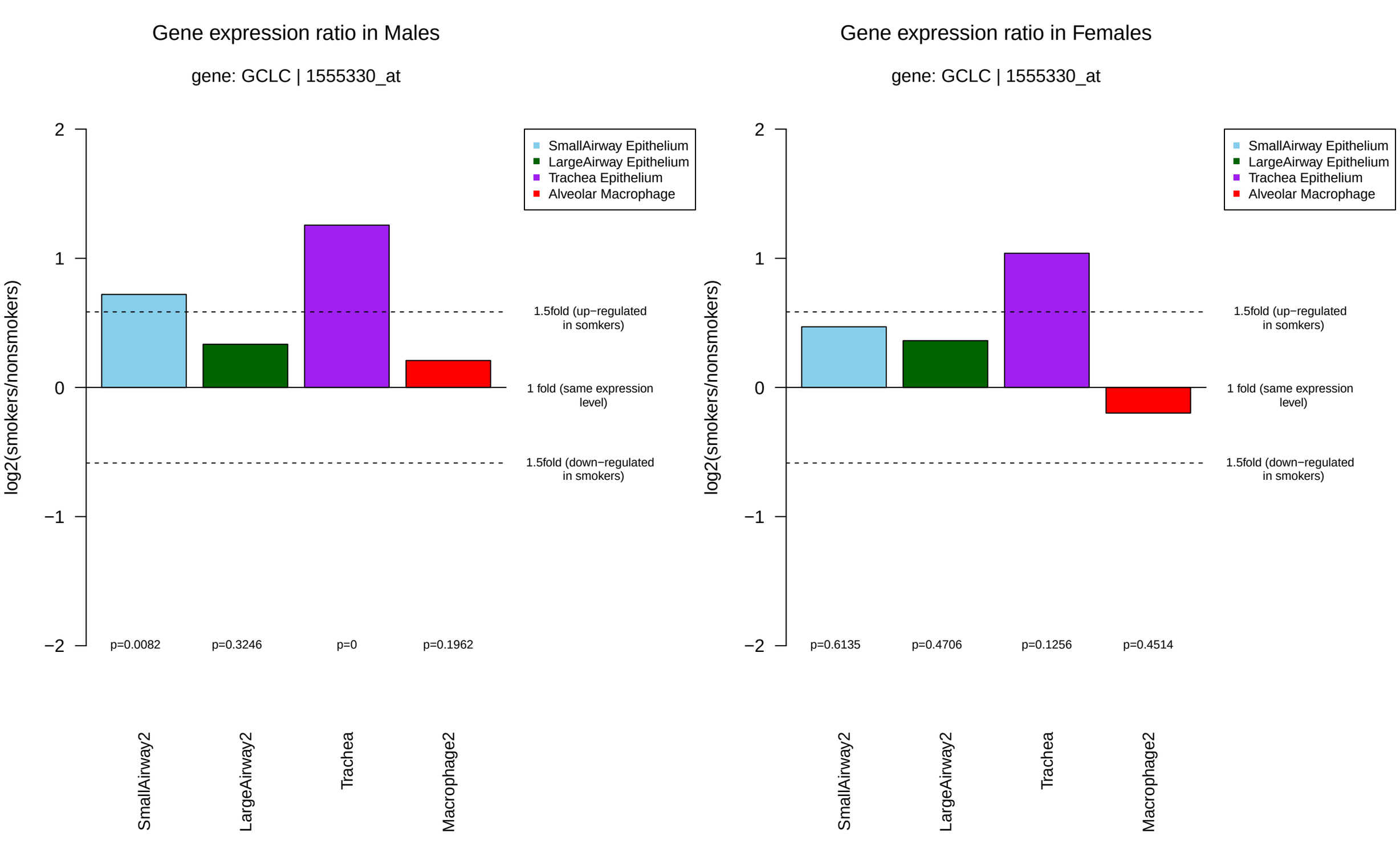
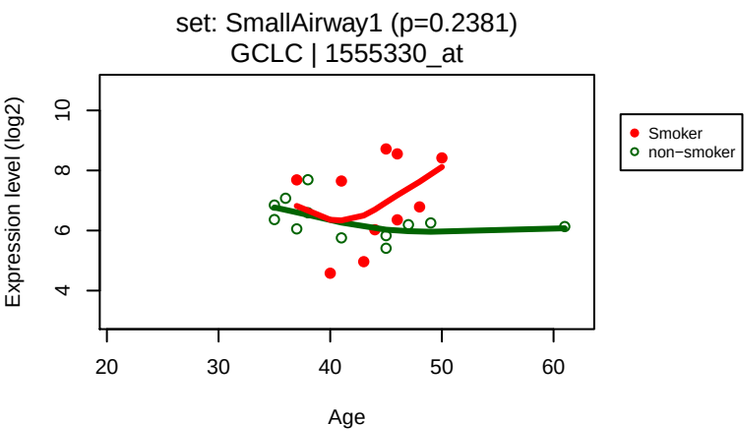
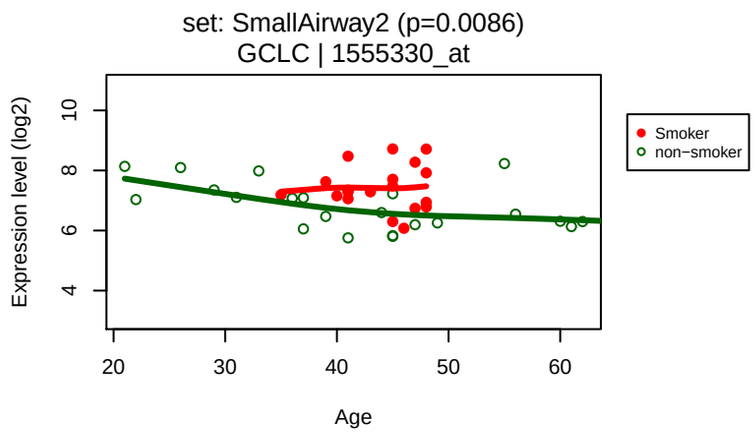
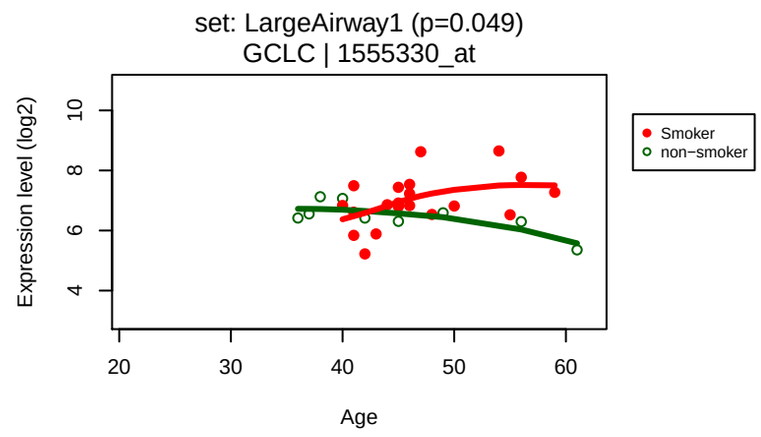

Gene expression of smokers and nonsmokers
gene: GCLC | 1555330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2381
p=0.0086
p=0
p=0.049
p=0.3403
p=0
p=0.0237
p=0.5819
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

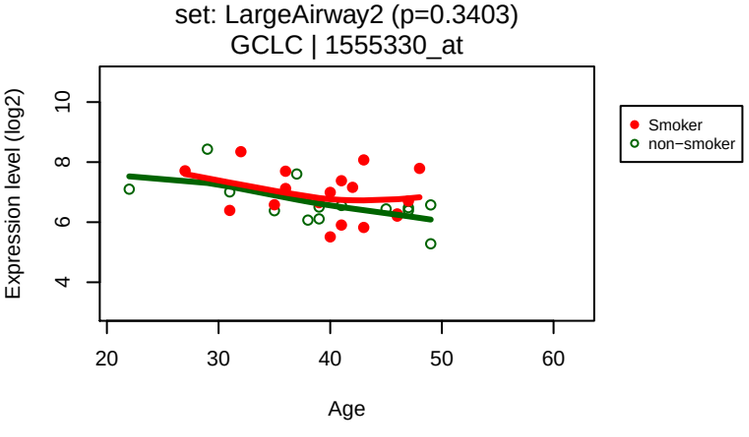

Gene expression ratio (smokers/nonsmokers)
gene: GCLC | 1555330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2381
p=0.0086
p=0
p=0.049
p=0.3403
p=0
p=0.0237
p=0.5819
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

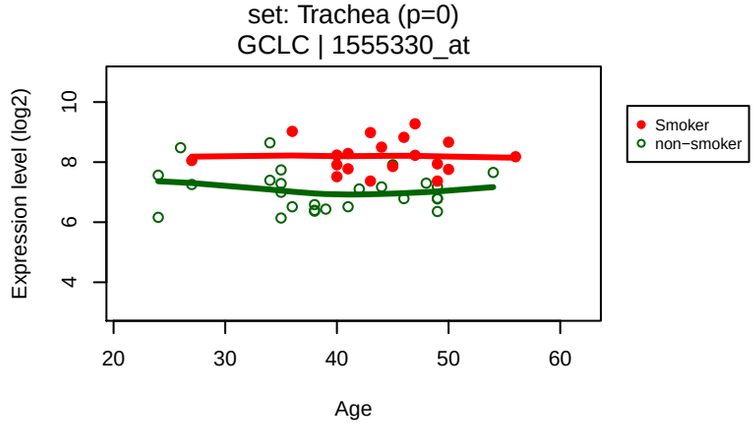

Gene expression ratio in Males
gene: GCLC | 1555330_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0082
p=0.3246
p=0
p=0.1962
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCLC | 1555330_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6135
p=0.4706
p=0.1256
p=0.4514
SmallAirway2
LargeAirway2
Trachea
Macrophage2

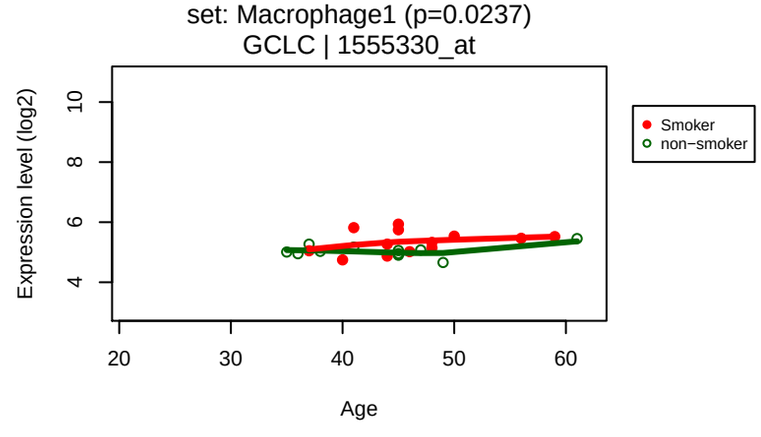

set: SmallAirway1 (p=0.2381)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker

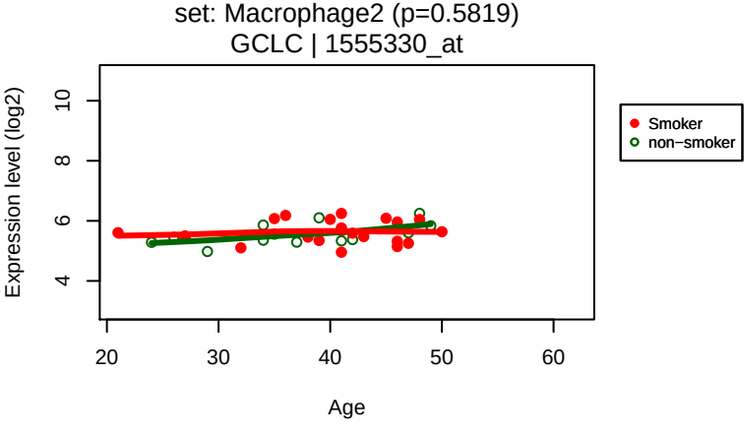

set: SmallAirway2 (p=0.0086)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker

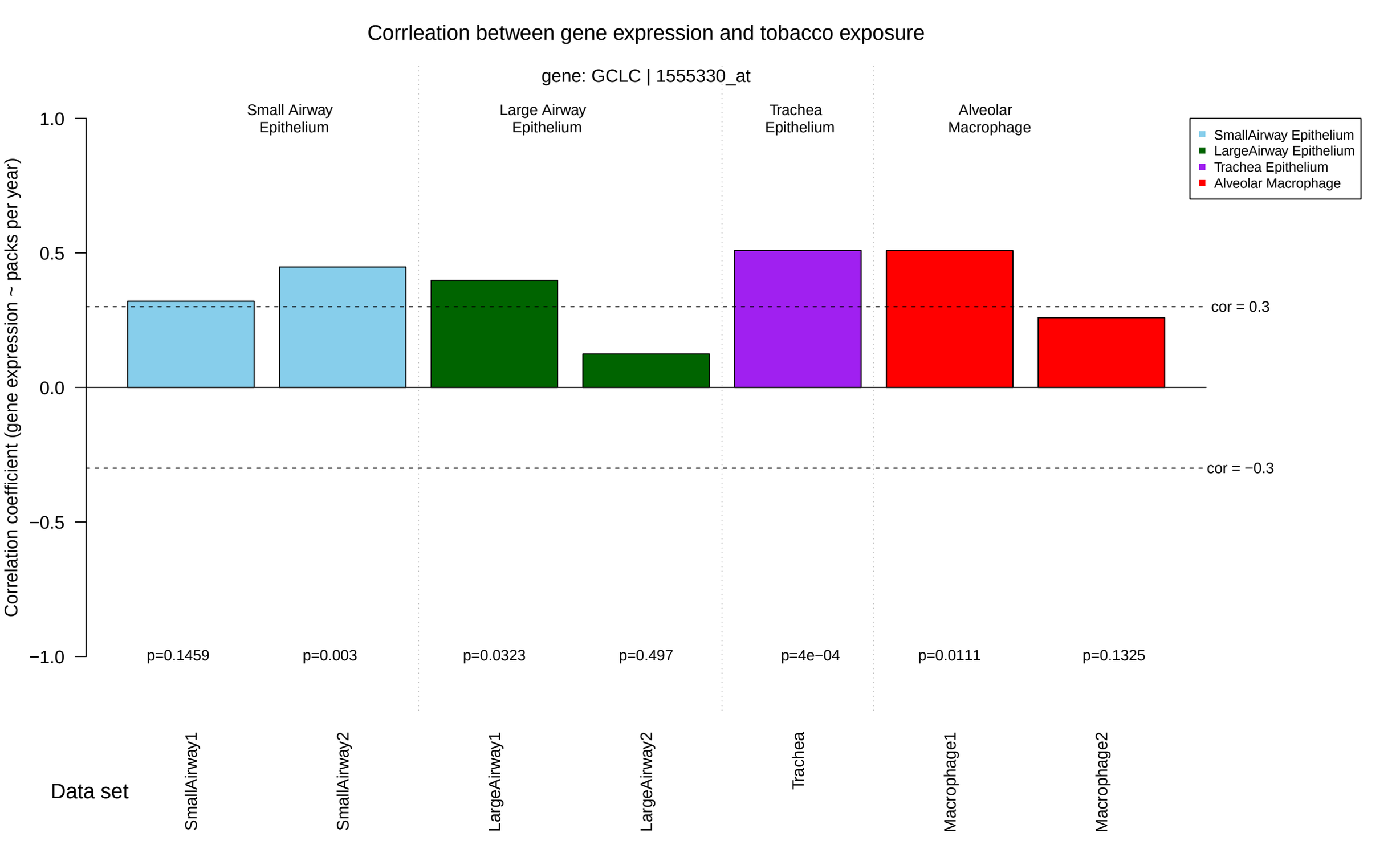

set: LargeAirway1 (p=0.049)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3403)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0237)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5819)
GCLC | 1555330_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCLC | 1555330_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1459
p=0.003
p=0.0323
p=0.497
p=4e−04
p=0.0111
p=0.1325
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2